THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET B

Full Syllabus (Chapters 9–14) | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The statements “If X then Y” and “X implies Y” have: 1

(a) the same meaning **(b)** opposite meanings

(c) unrelated meanings **(d)** the meaning of the converse

2. **Assertion (A):** The converse of the Baudhāyana–Pythagoras theorem is true. 1

Reason (R): If $a^2 + b^2 = c^2$, a right triangle with legs a, b has hypotenuse c, so the given triangle is congruent to it by SSS.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

3. How much water must be added to 9 litres of a 20% solution to reduce its concentration to 15%? 1

(a) 4 litres **(b)** 5 litres **(c)** 6 litres **(d)** 3 litres

4. 150 mL of a solution with 30% salt is mixed with 350 mL of a solution with 10% salt. The concentration of salt in the mixture is: 1

(a) 18% **(b)** 17% **(c)** 16% **(d)** 15%

5. By Euclid's algorithm, gcd(375, 825) is: 1

(a) 76 **(b)** 75 **(c)** 150 **(d)** 37

6. Euclid's subtraction algorithm is based on the fact that for m ≥ n, gcd(m, n) equals: 1

(a) gcd(m − n, m + n) **(b)** gcd(m × n, n)

(c) gcd(m, m + n) **(d)** gcd(n, m − n)

7. In quadrilateral ABCD, the sides opposite to AB and the angle opposite to ∠A are respectively: 1

(a) AD and ∠D **(b)** CD and ∠C **(c)** BC and ∠B **(d)** CD and ∠B

8. In △ABC, P and Q are the midpoints of AB and AC and BC = 9 cm. The length of PQ is: 1

(a) 18 cm **(b)** 9 cm **(c)** 6.5 cm **(d)** 4.5 cm

9. The solution of the pair x + y = 5 and x − y = 1 is: 1

(a) (-3, 2) **(b)** (3, 2) **(c)** (2, 3) **(d)** (3, -2)

10. A pair of linear equations has no solution when their graphs are: 1

(a) intersecting lines **(b)** perpendicular lines only

(c) parallel lines **(d)** coincident lines

11. The surface area of a cylindrical tumbler closed at one end and open at the other (radius r, height h) is: 1

(a) $\pi r(2h + r)$ **(b)** $\pi r(h + 2r)$ **(c)** $2\pi r(h + r)$ **(d)** $2\pi rh$

12. If the radius of a sphere is doubled, its volume becomes: 1

(a) 8 times **(b)** 2 times **(c)** 4 times **(d)** 6 times

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. Show that the number 9 is a counterexample to the proposition "If a number is odd, then it is prime". 2

14. Why is the final carry written in Step 5 of the addition algorithm? Give an example in which omitting it gives a wrong answer. 2

15. G is the centroid of $\triangle ABC$ and AD is a median of length 21 cm. Find AG and GD. 2

16. Find the slope of the line through $(-2, 5)$ and $(4, -1)$. 2

17. Find the curved surface area of a cylinder of radius 7 cm and height 20 cm ($\pi = \frac{22}{7}$). 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

18. Write the converse of "If a number is a multiple of 6, then it is a multiple of 3". Justify the proposition and give a counterexample for the converse. 3

19. Prove that if the opposite angles of a quadrilateral are equal, it is a parallelogram. 3

20. Solve by the elimination method: $3x + 4y = 18$ and $5x - 2y = 4$. 3

21. A conical cup of height 18 cm is filled with water up to a depth of 6 cm (measured from the apex). What fraction of the cup's volume is occupied by water? 3

22. A father is 4 times as old as his son. After 5 years, he will be 3 times as old as his son. Find their present ages. 3

OR

3 kg apples and 2 kg grapes cost ₹540, while 2 kg apples and 5 kg grapes cost ₹690. Find the price per kg of each fruit. 3

23. A cube of side 4 cm is painted on all faces and then cut into 64 cubes of side 1 cm. How many small cubes have 3

(i) three faces painted,
(ii) two faces painted,
(iii) one face painted,
(iv) no face painted?

OR

A metal sphere of radius 3 cm is melted and recast into small cones of radius 1 cm and height 3 cm. How many cones are made? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. Show that the triangle with vertices on the grid $A(0, 0)$, $B(7, 0)$, $C(0, 24)$ is right-angled using the Baudhāyana–Pythagoras converse, and find its hypotenuse and area. 5

OR

(i) Show that a perfect square has an odd number of factors by pairing factors, using the number 49 and the number 100. 5
(ii) Write the converse and say whether it is true.

25. Describe the three successive improvements to the first gcd algorithm (Section 11.3.1) and how many numbers are checked after each for m = 36, n = 60. 5

OR

Write the complete first algorithm for gcd(m, n) using lists of divisors. Execute it for gcd(30, 45). 5

END OF QUESTION PAPER Total: 50 Marks